

Before starting, make sure you **read all the questions** and understand what is asked of you. Write your answers to the questions in a separate sheet of paper and make sure you **justify all of your answers**. Multiple-choice questions have a **unique correct answer**.

For the questions below, we will focus on the language of (simplified) JSON documents. A JSON document is a **value**. In turn, a value can be one of the following:

- an **object**,
- an **array**,
- a non-empty string (made up of alphabetic symbols only, unquoted),
- a (possibly negative) integer number,
- a constant from true, false, or null.

Then, an object is a curly-brace surrounded, comma-separated sequence of pairs $k : v$ where k is a **key** (string) and v is a value. An array is a comma-separated sequence of values surrounded by square braces.

1. (5 points) Is the language of JSON documents a regular language? Why or why not?
2. (5 points) Write regular expressions for lexical units string, for number, and for all three constants.
3. (5 points) Give a context free grammar for the language of JSON documents.
4. (10 points) Suppose we are given a **context free grammar** for a language $\mathcal{L}$ on Σ . For all words $w \in \Sigma^*$, we can use the grammar to determine whether $w \in \mathcal{L}$ in time:
 - A. polynomial in $|w|$
 - B. exponential in $|w|$
 - C. nope, that is an undecidable problem
5. (10 points) Compute $\text{First}^2(\text{Atom})$ and $\text{Follow}^2(\text{Prod})$ based on the following grammar.

(1)	S	$\rightarrow$	$\text{Exp}\$$
(2)	Exp	$\rightarrow$	$\text{Prod Exp}'$
(3)	Exp'	$\rightarrow$	$+\text{Prod Exp}'$
(4)		$\rightarrow$	$-\text{Prod Exp}'$
(5)		$\rightarrow$	ε
(6)	Prod	$\rightarrow$	$\text{Atom Prod}'$
(7)	Prod'	$\rightarrow$	$*\text{Atom Prod}'$
(8)		$\rightarrow$	$/\text{Atom Prod}'$
(9)		$\rightarrow$	ε
(10)	Atom	$\rightarrow$	$-\text{Atom}$
(11)		$\rightarrow$	cst
(12)		$\rightarrow$	id
(13)		$\rightarrow$	(Exp)

6. (10 points) Construct the canonical finite-state machine for the following grammar with terminal set $T = \{a, b, c, d, \$\}$.

(1)	S	$\rightarrow$	$A\$$
(2)	A	$\rightarrow$	$aCbD$
(3)		$\rightarrow$	ab
(4)	C	$\rightarrow$	c
(5)	D	$\rightarrow$	d

7. (10 points) Is the **language** from the previous exercise LALR(1)?

8. (5 points) Which of the following is **not true** about the LLVM intermediate representation language?
- A. The number of registers is unlimited
 - B. The code has to be in SSA: single static assignment
 - C. Basic blocks can end without a terminator
 - D. Jumps can only target the start of a basic block
9. (10 points) Let $T = \{\text{id}, +\}$ be a set of terminals. Is the following grammar LL(1)? Does it define an LL(2) **language**?

(1)	S	$\rightarrow$	E
(2)	E	$\rightarrow$	$E + E$
(3)		$\rightarrow$	id

10. (10 points) During the course we have focused on conservative approximations of both type checking and reachability analysis. Argue that the exact versions of both tasks are in fact undecidable.
11. (10 points) Give an example of an LL(2) grammar that is not strongly LL(2).
12. (10 points) Give an example of a context-free language that is not LR(1).